

7.3 More while loop examples ⚙️ ^

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Beta features

Stepping through a while loop

The following program animation provides another while loop example. First, the user enters an integer at a time starting from the right, using % 10 to get the rightmost digit and // 10 to remove the rightmost digit. If the digit is greater than 0. Once num reaches 0, the loop will have printed all digits.

PARTICIPATION
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7.3.1: While loop step-by-step.

Start

☐ 2x speed

```
...
# Read num from user ...

# Print each digit
while num > 0:
    print(num % 10)
    num = num // 10
```

Iteration	num	Output
	902	
1		2
	90	
2		0
	9	
3		9
	0	

Printing out every digit of a user-entered number using the % and // operators.

Captions ^

1. User enters the number 902. The first iteration prints "2".

2. The second iteration prints "0".
3. The third iteration prints "9", so every digit has been printed. The loop condition is checked and the loop stops.

**PARTICIPATION
ACTIVITY****7.3.2: Stepping through a while loop.**

What is the output of the following blocks of code?

1)

```
count = 567
while count > 0:
    print(count % 10)
    count //= 10
```

- ☐ 7
- ☐ 6
- ☐ 5
- ☐ 6
- ☐ 5
- ☐ 7
- ☐ 5
- ☐ 6
- ☐ 7

2)

```
num = -123
while num > 0:
    print(num % 10 * 2)
    num //= 10
```

- ☐ -1
- ☐ -2
- ☐ -3
- ☐ -6
- ☐ -4
- ☐ -2
- ☐ No output is printed

3)

```
color = "purple"
n = 1
while n < len(color):
    print(color[n])
    n += 2
```

- ☐ u
- ☐ p
- ☐ e
- ☐ p
- ☐ r
- ☐ l
- ☐ IndexError

Greatest common divisor

The following program uses a while loop to compute the greatest common divisor (GCD) among two numbers, `num_a` and `num_b`, using Euclid's algorithm:

- If `num_a > num_b`, set `num_a` to `num_a - num_b`
- Else set `num_b` to `num_b - num_a`
- Repeat until `num_a` equals `num_b`, at which point `num_a` and `num_b` both equal the GCD

Try 7.3.1: While loop example: GCD program.

Try running the program below that calculates the GCD of two positive integers.



▶ Run

```
1  num_a = int(input("Enter first positive integer: "))
2  print()
3
4  num_b = int(input("Enter second positive integer: "))
5  print()
6
7  while num_a != num_b:
8      if num_a > num_b:
9          num_a = num_a - num_b
10     else:
11         num_b = num_b - num_a
12
13  print(f"GCD is {num_a}")
```

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7.3.3: Loop example: Greatest common divisor.

Use input values of **num_a = 15** and **num_b = 10** in the above GCD program. Answer the questions. If stuck, consider adding additional print statements to the program.

1) What is the value of num_a before the first loop iteration?

Check

Show answer

2) What is num_a after the first iteration and before the second?

Check

Show answer

3) What is num_b after the second and before the third iteration?

Check

Show answer

4) How many loop iterations will the algorithm execute?

Check

Show answer

Conversation

Below is a program that has a "conversation" with the user. The program asks the user to type some four possible responses until the user enters "Goodbye". Each time through the while loop, the program checks if user_text is equal to the string literal "Goodbye". If the two strings are not equal, the while loop body

Within the while loop body, the program obtains a random number between 0 and 2 by using the randint() function. The function provides a new random number each time the function is called. The arguments to randint(), 0 and 2, are the minimum and maximum values that the function may return. Using the number given by randint(), the program's elif statements

Note that the first few lines of the program represent a **docstring**: a multi-line string literal delimited by triple quotes, which can be either single (') or double (") quotes.

Try 7.3.2: While loop example: Conversation.

Run the program below. Try adding additional conditions to leave the conversation, such as "See with the string "Goodbye".



▶ Run

```
1  """
2  Program that has a conversation with the user.
3  Uses elif branching and a random number to mix up the program's responses.
4  """
5
6  import random # Import a library to generate random numbers
7
8  print("Tell me something about yourself.")
9  print("You can type \"Goodbye\" at anytime to quit.\n")
10
11 user_text = input()
12
13 while user_text != "Goodbye":
14     random_num = random.randint(0, 2) # Gives a random integer between 0 and
15     if random_num == 0:
16         print("\nPlease explain further.\n")
17     elif random_num == 1:
18         print(f"\nWhy do you say: \"{user_text}\"?\n")
19     elif random_num == 2:
20         print("\nWhat else can you share?\n")
21     else:
22         print("\nUh-oh. something went wrong. Try again.\n")
```

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Model Solution ▼

**PARTICIPATION
ACTIVITY****7.3.4: Conversation.**

Refer to the above program.

1) Which if-else branch will execute if the user types "Goodbye"?

- ☐ `if random_num == 0:`
- ☐ `else`
- ☐ `none`

2) How many times does the loop iterate in the program?

- ☐ 0
- ☐ 2
- ☐ unknown

3) Which of the following is an expression using `random.randint()` that returns a number between 0 and 5, inclusive?

- ☐ `random.randint(-1, 6)`
- ☐ `random.randint(0, 5)`
- ☐ `random.randint(0, 6)`

Example: Getting input until a sentinel is seen

Loops are commonly used to process a series of input values. A sentinel value is used to terminate computes the average of an input list of positive integers, ending with 0. The 0 is not included in the

Try 7.3.3: Computing average of a list with a sentinel.



▶ Run

```
1  """
2  Outputs average of list of positive integers
3  List ends with 0 (sentinel)
4  Ex: 10 1 6 3 0  yields (10 + 1 + 6 + 3) / 4, or 5
5  """
6
7  values_sum = 0
8  num_values = 0
9
10 curr_value = int(input())
11
12 while curr_value > 0: # Get values until 0 (or less)
13     values_sum += curr_value
14     num_values += 1
15     curr_value = int(input())
16
17 print(f"Average: {values_sum / num_values:.0f}\n")
18
```

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7.3.5: Average example with a sentinel.

Consider the example above and the given example input sequence 10 1 6 3 0.

- 1) How many actual (non-sentinel) values are given in the first input sequence?
 - ☐ 1
 - ☐ 4
 - ☐ 5
- 2) For the given input sequence, what is the final value of num_values?
 - ☐ 0
 - ☐ 4
 - ☐ 5
- 3) Suppose the first input was 0. Would values_sum / num_values be 0?
 - ☐ Yes
 - ☐ No
- 4) What would happen if the following list was input: 10 1 6 3 -1?
 - ☐ Output would be 5
 - ☐ Output would be 4
 - ☐ Error

**CHALLENGE
ACTIVITY****7.3.1: While loop with sentinel.**

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Start

Type the program's output

```
curr_value = int(input())  
  
max_value = curr_value  
  
while curr_value > 0:  
    if curr_value > max_value:  
        max_value = curr_value  
    curr_value = int(input())  
  
print(f"Max value: {max_value}")
```

Input

6
7
4
9
2
0

Output

Max value: 9

1

2

3

Check**Next**View solution ▼ *(Instructors only)***CHALLENGE
ACTIVITY**

7.3.2: Bidding example.

763650.1722066.qx3zqy7

Drag and drop the correct line of code that causes the while loop to continue bidding until the

Note: Only one line of code on the left should be used in the final solution.

How to use this tool 

Unused

```
while keep_bidding != "y":
```

```
while keep_bidding == "n":
```

```
while keep_bidding = "y":
```

```
while keep_bidding != "n":
```

main.py

```
import random
random.seed(5)
```

```
keep_bidding = "y"
next_bid = 0
```

```
next_bid = next_bid +
print(f"I'll bid ${ne
print("Continue biddi
keep_bidding = input(
```

Check

View solution  (Instructors only)

CHALLENGE ACTIVITY

7.3.3: While loop: Insect growth.

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Positive integer `num_insects` is read from input. Organize the lines of code to create a while loop is less than or equal to 100. In each iteration:

- Print `num_insects`.
- Update `num_insects` with `num_insects` multiplied by 2.

Ex: If the input is 8, then the output is:

8
16
32
64

How to use this tool ▼

Unused

```
print(num_insects)
while num_insects <= 100:
    num_insects = num_insects * 2
```

main.py

```
num_insects = int(input())
```

Check

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CHALLENGE ACTIVITY

7.3.4: Advanced while loop examples.

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Start

Integer `basque_count` is initialized with 0. String `input_item` is read from input. Write a loop that equal to "Stop". In each iteration of the loop:

- Increment `basque_count` if `input_item` is equal to "Basque".
- Read string `input_item` from input.

[Click here for example](#) ▼

```
1 basque_count = 0
2
3 input_item = input()
4
5 while "":
6     "":
7
8
9 print(f"Basque occurs {basque_count} time(s).")
```

1

2

Check**Next level**

View solution ▼ *(Instructors only)*